

Rajat Kumar Saini

9760685744 | rajatsaini798317@gmail.com | [LinkedIn](#)

SUMMARY

Node.js Developer with 2+ years of experience designing and developing scalable backend systems, RESTful APIs, and microservices using Node.js, Express, and NestJS. Expertise in real-time systems with WebSockets, RabbitMQ, and Kafka for event-driven architecture. Proficient in JavaScript/TypeScript, MongoDB, MySQL, PostgreSQL, and third-party API integration. Hands-on with AWS, Docker, CI/CD, and high-load performance optimization. Built production systems supporting 20,000+ active users with HLS streaming, RBAC, and asynchronous queue processing. Strong problem-solving, debugging, and ability to take ownership.

EXPERIENCE

Backend Developer (Node.js)

February 2025 – Present

Aucourant Technologies Pvt.

Dehradun, Uttarakhand

- Designing and developing scalable backend systems using Node.js, Express for production environments.
- Building RESTful APIs and microservices for real-time video streaming, activity detection, and third-party service integration.
- Working on real-time features using WebSockets and event-driven systems with RabbitMQ for asynchronous processing.
- Integrating third-party APIs and services, including AWS S3, HLS video streaming, and payment gateways.
- Optimized backend performance for high-load systems by reducing database query latency by 40% and supporting 20,000+ active users
- Collaborating with frontend and product teams in Git-based workflows with code reviews and CI/CD pipelines.

Node.js Backend Developer

January 2024 – February 2025

MYC India Education Pvt.

Mohali, Punjab

- Developed scalable backend services and REST APIs using Node.js, Express.js, and Sequelize ORM.
- Managed databases including MongoDB, MySQL with optimized indexing and query performance
- Contributed to scalable application architecture decisions, including API versioning, modular service design, and error-handling middleware.

Internships, Software Developer

June 2022 - December 2022

Jsimple Technosoft Pvt.

Noida sector 62

- Contributed to ASP.NET MVC web applications, improving UI workflows and reducing page load times.
- Authored stored procedures and database views to streamline data access and reduce redundant query logic.
- Participated in peer code reviews and sprint-based feature delivery.

PROJECTS

Cratch — AI-Based Child Activity Monitoring System | Node.js, Python, OpenCV, MongoDB Cratch

- Designed and developed a real-time child activity monitoring platform using a microservice architecture: Node.js handles REST API, WebSocket event brokering, and MongoDB storage; Python (OpenCV) handles AI-based video analysis.
- Implemented live video stream ingestion and frame-level activity detection (eating, reading, sleeping) using OpenCV, with classified events stored asynchronously in MongoDB for audit trails.
- Enabled real-time bidirectional communication via WebSockets, delivering activity alerts to parent dashboards with sub-second latency.
- Built an event-driven system for behavioral pattern tracking with configurable alert thresholds and notification dispatch.
- Enabled live video streaming and activity tracking through WebSockets.

SELMS (LMS Platform) | MongoDB, Express.js, React.js, Node.js, AWS S3, RabbitMQ SELMS

- Engineered a production-grade Learning Management System scaled to 20,000+ active users, built on the MERN stack (MongoDB, Express.js, React.js, Node.js).
- Implemented AWS S3 integration with HLS video streaming for adaptive, bandwidth-efficient course content delivery.

- Architected an async job queue with RabbitMQ for background processing (email notifications, report generation, bulk imports), decoupling heavy workloads from synchronous API requests.
- Built an Excel-based bulk data import pipeline handling thousands of records per batch with validation and rollback support.
- Deployed automated email notification system for enrollment, progress, and certification milestones.
- Established CI/CD pipeline using GitHub Actions for automated testing, build, and zero-downtime deployment.
- Integrated Razorpay payment gateway with webhook handling for real-time payment status updates and subscription management.

CRM Web Application | Angular, Node.js, MySQL, Laravel

CRM

- Developed backend REST APIs and database workflows for a CRM system with MySQL, supporting lead tracking, contact management, and pipeline analytics.
- Built role-based dashboards with granular permission controls (RBAC) and an application tracking system with status-change event logging.
- Implemented real-time team notifications using WebSockets for instant CRM updates across concurrent users.

Study Abroad Platform | Angular, Node.js, MySQL

Study Abroad

- Built an eligibility evaluation engine assessing student profiles against exam score thresholds for university and course matching.
- Developed a full-text and filter-based search system for 1,000+ courses and universities with pagination and sorting.

EDUCATION

Uttarakhand Technical University <i>Master of Computer Applications</i>	Dehradun, India <i>August 2021 – May 2023</i>
Gurukul Kangri University <i>Bachelor of Science (Computer Science)</i>	Haridwar, India <i>August 2018 – May 2021</i>

TECHNICAL SKILLS

Languages: JavaScript (ES2022+), TypeScript, Python, SQL, HTML5, CSS3
Backend: Node.js, Express, NestJS, RESTful APIs, Microservices, WebSockets, API Design
Frontend: React.js, Angular, Bootstrap, Material UI, jQuery
Databases: MongoDB, MySQL, MS SQL Server, Sequelize ORM, Query Optimization, Indexing
Real-time/Queues: RabbitMQ, Redis (conceptual), Async Processing, Event-Driven Architecture
Cloud/DevOps: AWS S3, EC2, Lambda, CloudFront, Docker, CI/CD, Git, GitHub Actions
AI Integration: Python (OpenCV), Video Stream Processing, AI Microservice Integration, Activity Detection
Architecture: MVC, REST, Microservices, Scalable Backend Design, API Versioning, Performance Optimization